

SpaceX Falcon 9 First Stage Landing Prediction

GitHub Repository:

<https://github.com/Satish-KumarRam/IBM-Data-Science-Capstone-Project>

**IBM Data Science Professional Certificate
Capstone Project**

Satish Kumar Ram

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Outline

- Executive Summary
- Introduction
- Data Collection & Data Wrangling Methodology
- EDA & Data Visualization Methodology
- EDA with Visualization
- EDA with SQL Queries
- Folium Map
- Plotly Dash Dashboard
- Predictive Analysis (Classification)
- Predictive Analysis Results
- Conclusion
- Appendix



Introduction

- **Project Background and Context:**

- SpaceX has significantly reduced the cost of space missions by successfully recovering and reusing Falcon 9 first-stage boosters. Traditional rocket launches are expensive because rockets are usually discarded after launch. By predicting whether the Falcon 9 first stage will land successfully, organizations can estimate mission costs, improve planning, and support data-driven decision-making. This project uses historical SpaceX launch data to analyze landing outcomes and build machine learning models for landing prediction.

- **Project Objective:**

- The project aims to identify the factors influencing Falcon 9 first-stage landing success and develop a machine learning model to predict landing outcomes.
- Collect Falcon 9 launch data using the SpaceX REST API and **Wikipedia Web Scraping**.
- Perform **data wrangling, SQL analysis, and exploratory data analysis (EDA)**.
- Build interactive visualizations using SQL, Folium, and Plotly Dash.
- Develop, evaluate, and compare machine learning classification models to predict Falcon 9 first-stage landing success.



Executive Summary

- **Objective**

- Predict Falcon 9 first-stage landing success using historical launch data and machine learning.

- **Summary of Methodology**

- Data Collection through **SpaceX REST API**.
- Data Collection using **Wikipedia Web Scraping**.
- Data Wrangling and Feature Engineering.
- Exploratory Data Analysis (EDA) using SQL Queries, Python Visualizations (Matplotlib & Seaborn), and Plotly Dash.
- Interactive Visual Analytics using **Folium** and **Plotly Dash**.
- Predictive Analysis using **Classification Models**.

- **Summary of Results**

- Landing success rate improved significantly over time.
- Payload mass, orbit type, and launch site influenced landing success.
- SQL and visualizations revealed important launch patterns.
- Interactive dashboards provided deeper insights into launch data.
- The best classification model achieved the highest prediction accuracy.



Methodology

- **Data Collection Methodology**

- Collected Falcon 9 launch data using the **SpaceX REST API**.
- Collected additional launch records through **Wikipedia Web Scraping**.

- **Data Wrangling**

- Cleaned missing values and standardized the dataset.
- Created the landing success label and prepared features for analysis.

- **Exploratory Data Analysis (EDA)**

- Performed **SQL queries** to analyze launch statistics.
- Created visualizations using Matplotlib and Seaborn.

- **Interactive Visual Analytics**

- Built an interactive **Folium** map to analyze launch site locations.
- Developed an interactive Plotly Dash dashboard with dropdown filters, charts, and data visualizations for launch analysis.

- **Predictive Analysis**

- Built, trained, and evaluated multiple machine learning classification models.
- Compared model performance and selected the best-performing model based on prediction accuracy.



Data Collection using SpaceX API

- Collected Falcon 9 launch data using the **SpaceX REST API**.
- Retrieved launch date, payload mass, orbit, launch site, booster version, and mission outcome.
- Converted JSON responses into a structured Pandas DataFrame for analysis.
- **Right side:** SpaceX API notebook screenshot.

```
import requests
import pandas as pd

static_json_url = "https://cf-courses-data.s3.us.cloud-object-storage.appdomain.cloud/IBM-DS0321EN-SkillsNetwork/datasets/API_call_spacex_api.json"

response = requests.get(static_json_url)

print(response.status_code)

200

We should see that the request was successful with the 200 status response code

response.status_code

200

Now we decode the response content as a Json using .json() and turn it into a Pandas dataframe using .json_normalize()

# Use json_normalize method to convert the json result into a dataframe
data = pd.json_normalize(response.json())

Using the dataframe data print the first 5 rows

# Get the head of the dataframe
data.head(5)
```

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Data Collection using Wikipedia Web Scrapping

- Extracted additional Falcon 9 launch records from **Wikipedia** using **BeautifulSoup**.
- Parsed HTML tables and converted them into a Pandas DataFrame.
- Used the scraped data to enrich and validate the dataset.
- **Right side:** Web Scrapping notebook screenshot.

```
# use requests.get() method with the provided static_url and headers
# assign the response to a object
response = requests.get(static_url, headers=headers)
```

Create a BeautifulSoup object from the HTML response

```
# Use BeautifulSoup() to create a BeautifulSoup object from a response text content

soup = BeautifulSoup(response.text, "html.parser")

print(soup.title.string)
```

List of Falcon 9 and Falcon Heavy launches - Wikipedia

Print the page title to verify if the BeautifulSoup object was created properly

```
# Use soup.title attribute
print(soup.title)

<title>List of Falcon 9 and Falcon Heavy launches - Wikipedia</title>
```

TASK 2: Extract all column/variable names from the HTML table header

Next, we want to collect all relevant column names from the HTML table header

Let's try to find all tables on the wiki page first. If you need to refresh your memory about BeautifulSoup, please check the external reference link towards the end of this lab

```
# Use the find_all function in the BeautifulSoup object, with element type `table`
# Assign the result to a list called `html_tables`

html_tables = soup.find_all('table')
```

Starting from the third table is our target table contains the actual launch records.

```
# Let's print the third table and check its content
first_launch_table = html_tables[2]
print(first_launch_table)
print(first_launch_table)
```

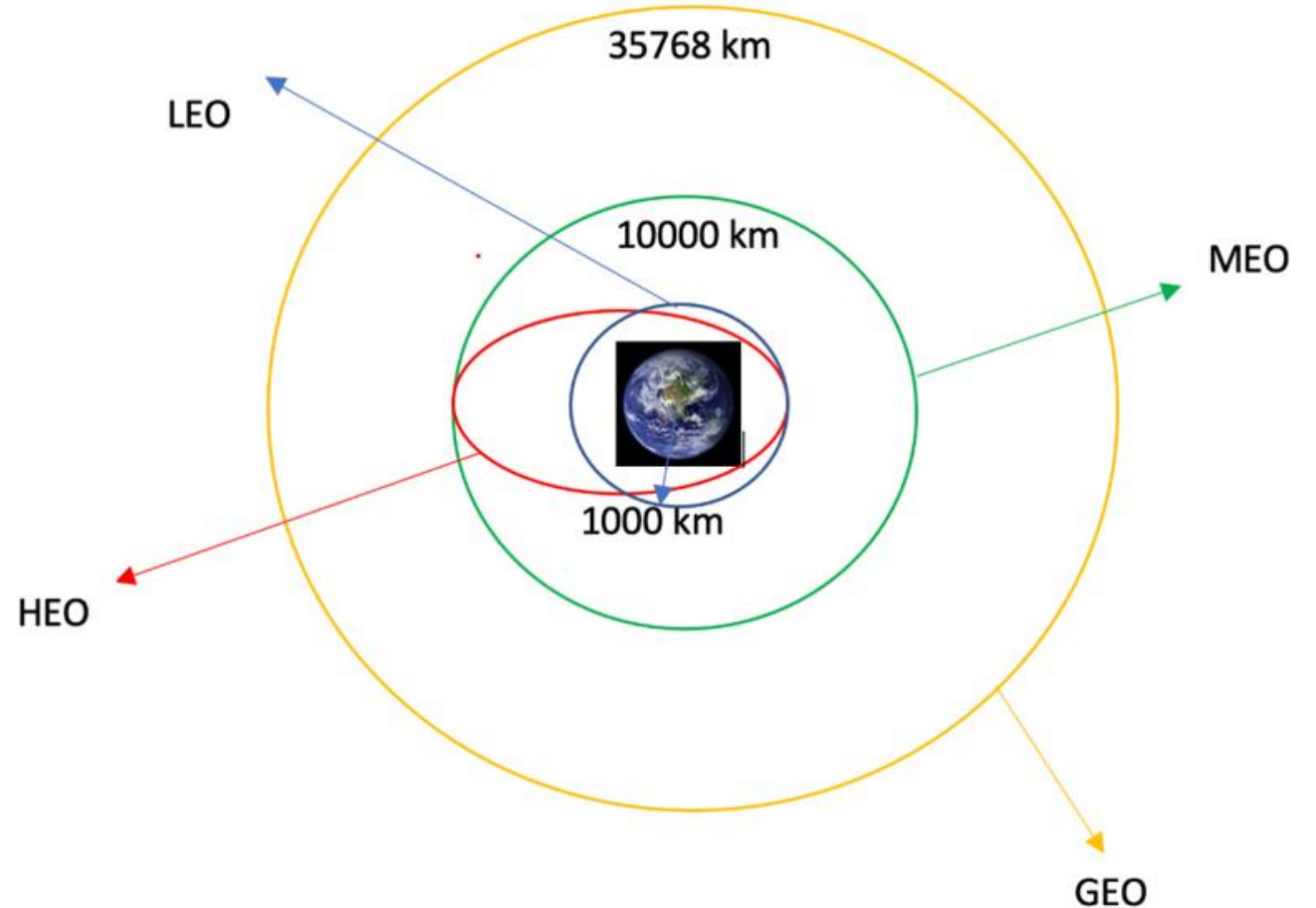


Data Wrangling

- Cleaned missing and inconsistent values.
- Selected relevant features for analysis.
- Created the landing success label.
- Prepared the final dataset for EDA and machine learning.

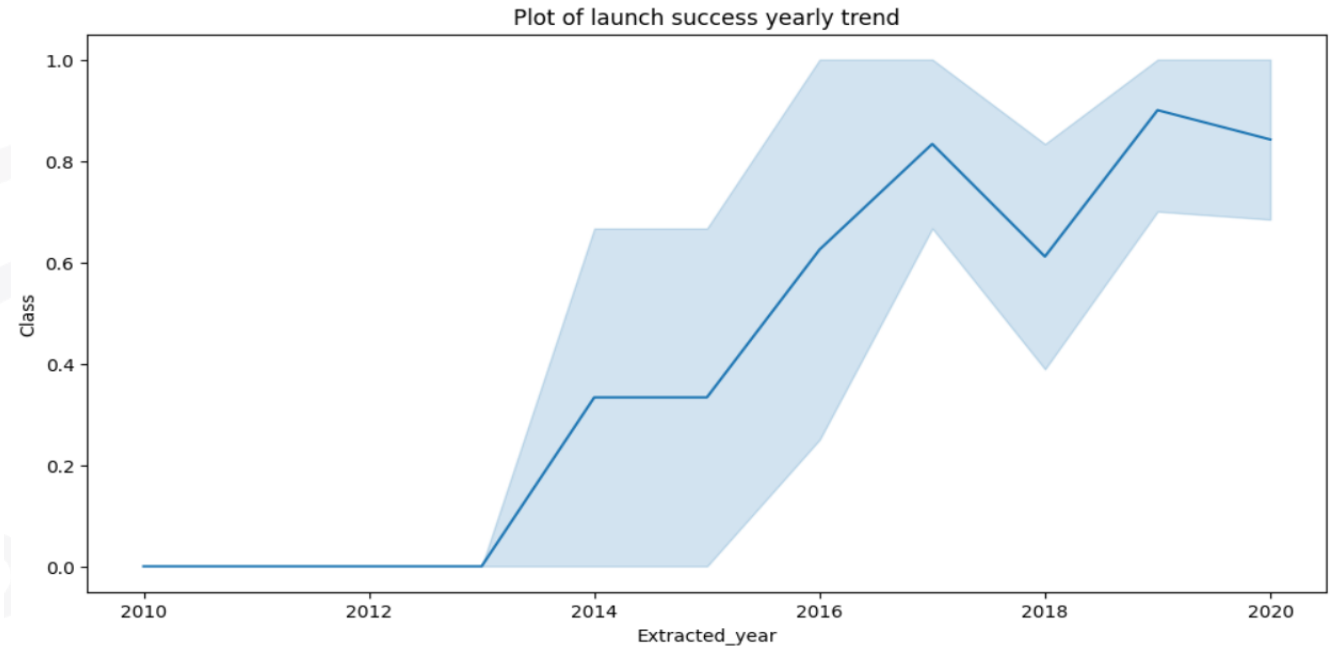
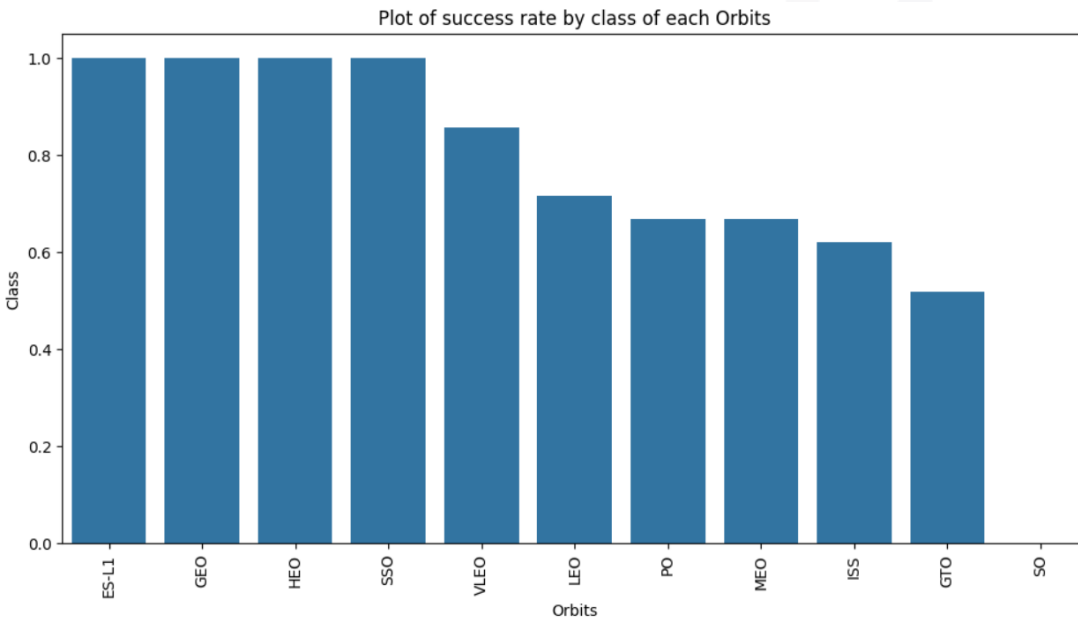
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EDA using Visualizations

- Launch success increased over time.
- Success rates differed across orbit classes.
- EDA revealed patterns influencing landing success.



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EDA using SQL

- Used SQL queries to summarize launch statistics.
- Analyzed launches by launch site, mission outcome, and booster version.
- Identified trends and generated insights from the data.

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Interactive Visual Analytics (Folium & Plotly Dash)

- Built interactive **Folium** maps to visualize launch site locations.
- Developed a **Plotly Dash** dashboard with filters and interactive charts.
- Enabled dynamic exploration of launch data.

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Predictive Analysis

- Built multiple classification models to predict landing success.
- Compared model performance using evaluation metrics.
- Selected the best-performing model based on prediction accuracy.

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EDA Using SQL Queries (Part 1)

SQL Query Output – Launch Site Analysis

```
%%sql
```

```
SELECT Launch_Site,  
COUNT(*) AS Total_Launches  
FROM SPACEXTABLE  
GROUP BY Launch_Site;
```

```
* sqlite:///my_data1.db  
Done.
```

Launch_Site	Total_Launches
CCAFS LC-40	26
CCAFS SLC-40	34
KSC LC-39A	25
VAFB SLC-4E	16

- **Key Findings:**
- Used SQL to calculate total launches for each Falcon 9 launch site.
- **CCAFS SLC-40** recorded the highest number of launches (**34**).
- **CCAFS LC-40** completed **26** launches during the analyzed period.
- KSC LC-39A completed 25 launches.
- **VAFB SLC-4E** recorded the fewest launches (**16**).
- These SQL results identified the busiest launch sites and supported subsequent visualization and predictive analysis.



EDA Using SQL Queries (Part 2)

Mission Outcome Analysis (SQL Query Result)

```
%%sql

SELECT Mission_Outcome,
       COUNT(*) AS Count
FROM SPACEXTABLE
WHERE Date BETWEEN '2010-06-04' AND '2017-03-20'
GROUP BY Mission_Outcome
ORDER BY Count DESC;

* sqlite:///my_data1.db
Done.
```

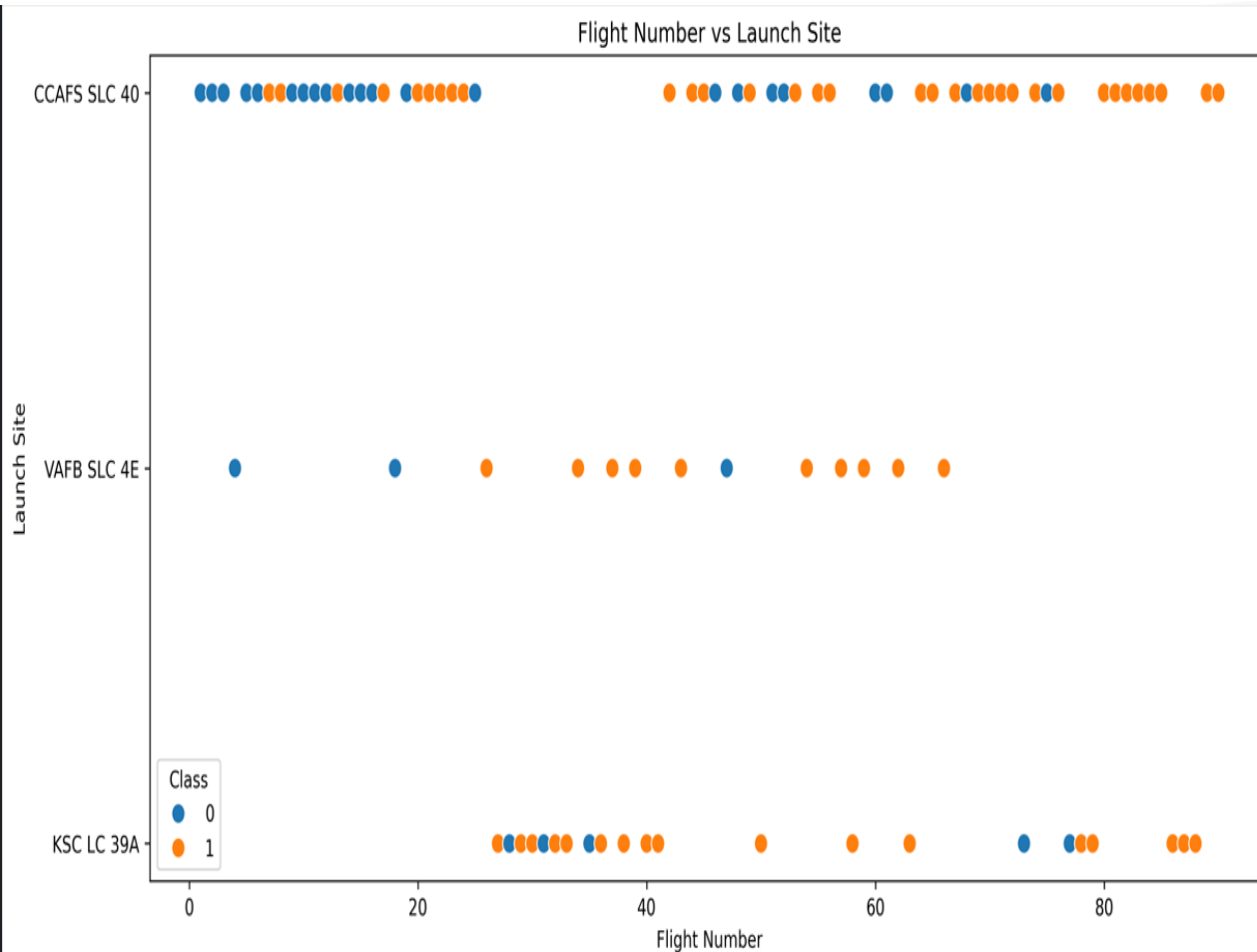
Mission_Outcome	Count
Success	30
Failure (in flight)	1

- **Key Findings:**
- SQL was used to summarize mission outcomes.
- **30 missions** were completed successfully.
- Only **1 mission** resulted in failure during the selected period.
- The high success rate indicates improved Falcon 9 reliability.
- SQL aggregation helped validate mission performance before predictive modeling.

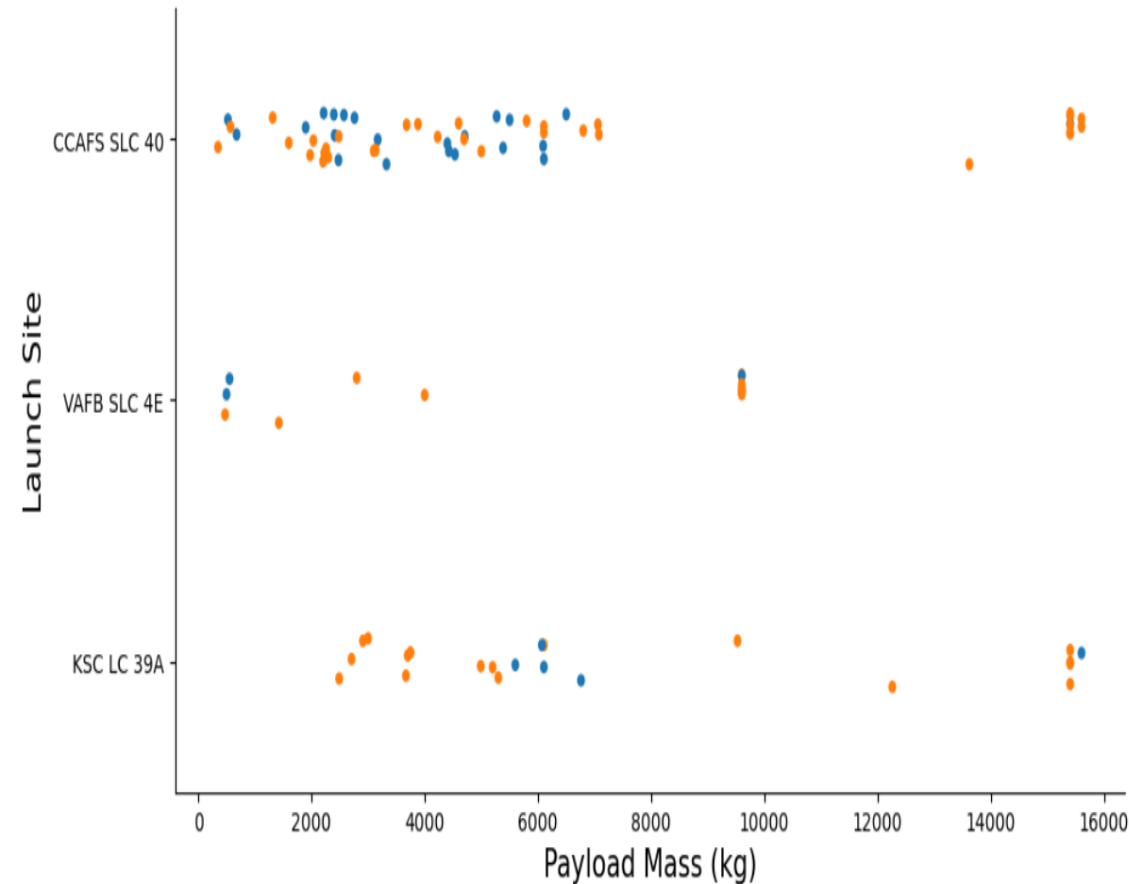
Results

- **Landing Success Trend:** This project integrates data collection, EDA, SQL, interactive dashboards, and machine learning to predict Falcon 9 landing success.
- **Key Factors:** Payload mass, launch site, orbit type, flight number, and booster version influenced landing success.
- **EDA, SQL & Visualization:** SQL queries, Folium maps, and the Plotly Dash dashboard provided interactive insights.
- **Predictive Analysis:** The best classification model achieved the highest prediction accuracy.
- **Overall Outcome:** Historical launch data can effectively predict Falcon 9 first-stage landing success and support mission planning.

Flight Number vs. Launch Site



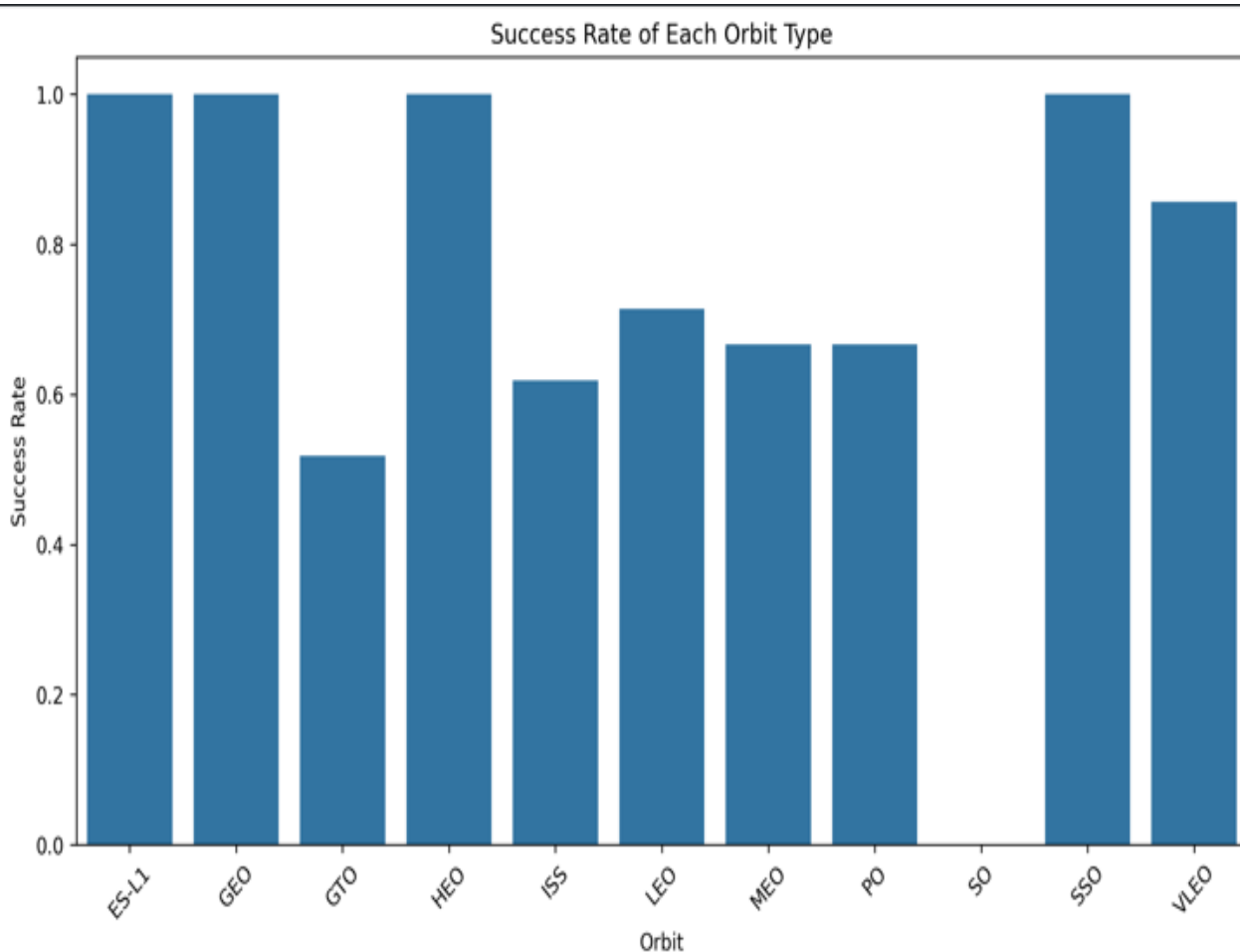
Payload Mass vs. Launch Site



- **Key Findings**

- **CCAFS SLC 40** handled the widest range of payload masses and was the most frequently used launch site.
- **KSC LC 39A** primarily handled medium- to high-payload missions and achieved several successful landings.
- **VAFB SLC 4E** conducted fewer launches but demonstrated successful missions across different payload categories.
- Both **successful** and **unsuccessful** landings occurred at low, medium, and high payload masses, indicating that **payload mass alone was not the primary factor affecting landing success**.
- Overall, launch site and payload mass together influenced Falcon 9 landing success.

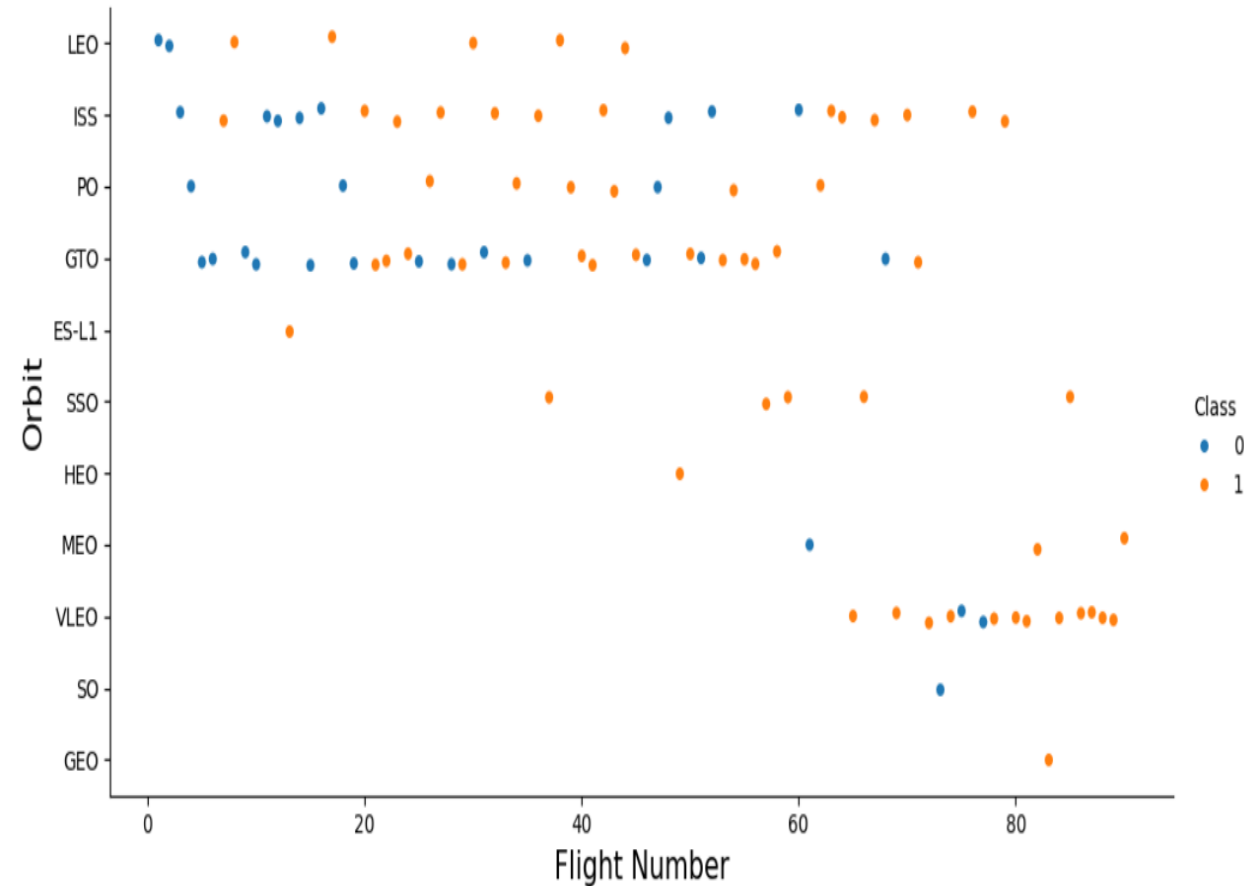
Success Rate by Orbit



• Key Findings

- **ES-L1, GEO, HEO, and SSO** achieved a **100% landing success rate**, indicating highly reliable mission outcomes for these orbit types.
- **VLEO** also demonstrated a high landing success rate of approximately **86%**, showing strong booster recovery performance.
- LEO, MEO, ISS, and PO achieved moderate landing success rates, indicating variable mission outcomes across these orbit types.
- **GTO** recorded a comparatively lower landing success rate, while **SO** showed **no successful landings** in the available dataset.
- Overall, orbit type significantly influenced Falcon 9 landing success, although mission characteristics and operational factors also played important roles.

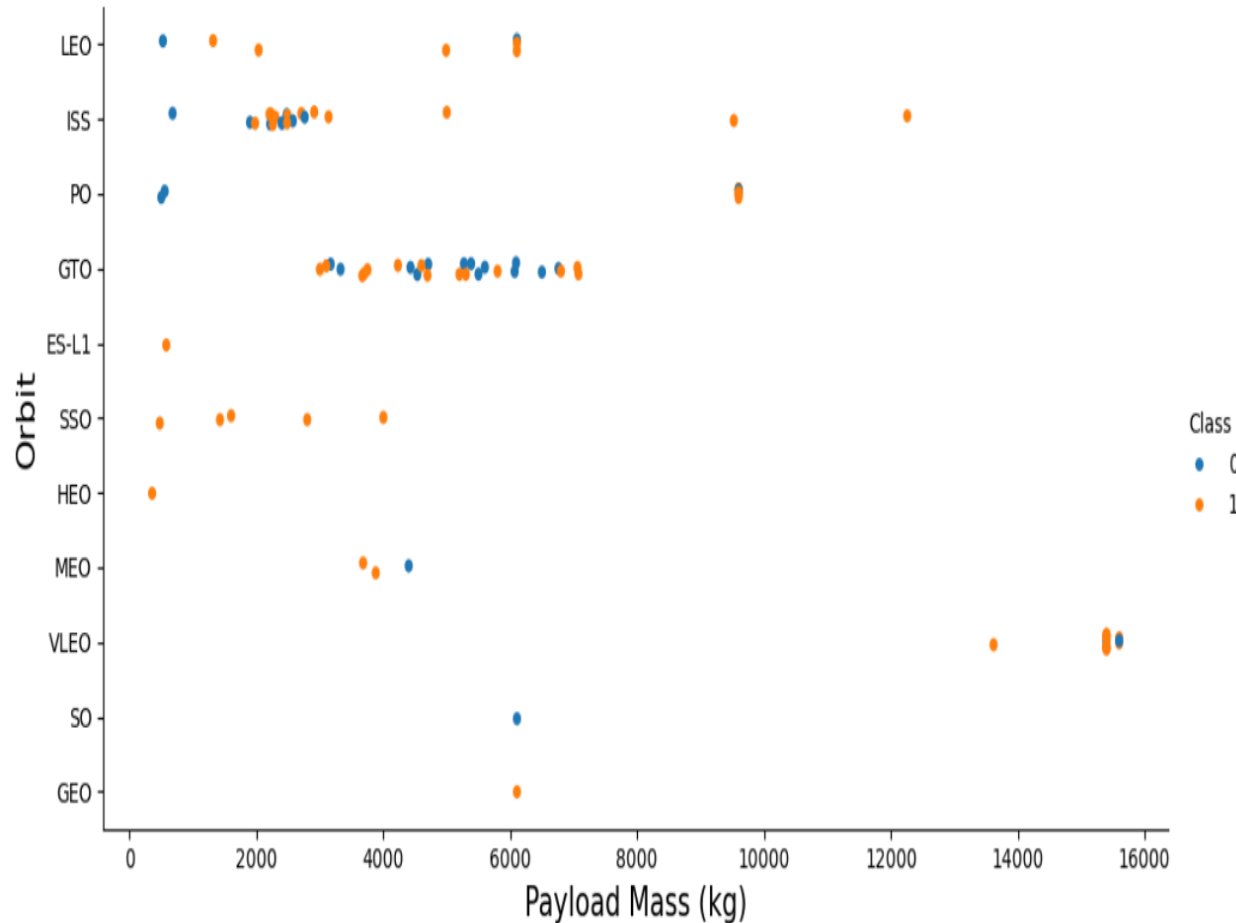
Flight Number vs. Orbit



• Key Findings

- LEO, ISS, and VLEO missions generally achieved high landing success rates.
- GTO missions showed mixed landing outcomes, with both successful and failed landings.
- Landing success generally improved with increasing flight numbers, especially in later missions.
- Orbit type influenced landing success, while payload mass and launch site also played important roles.

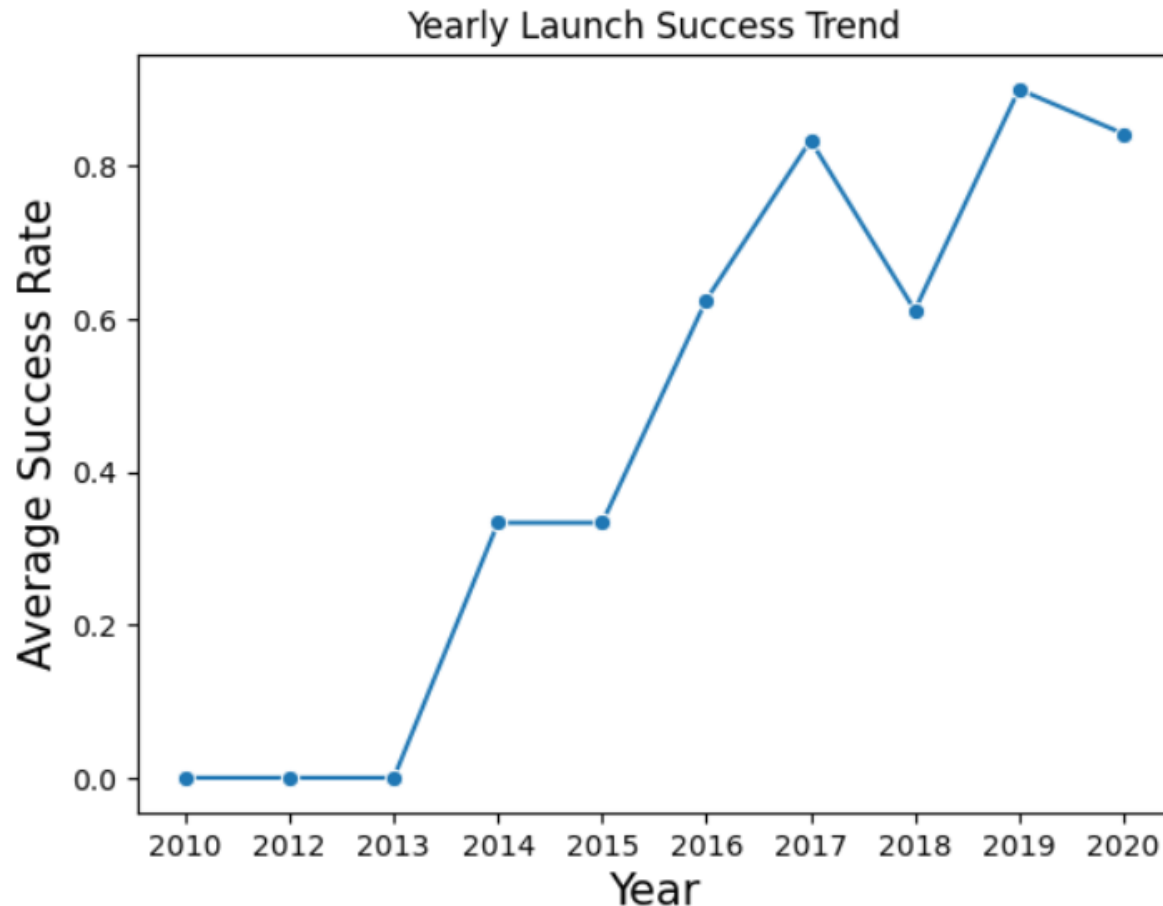
Payload Mass vs. Orbit



• Key Findings

- GTO missions covered a wide range of payload masses, with both successful and unsuccessful landings observed.
- VLEO missions were associated with the highest payload masses.
- LEO and ISS missions generally carried low to medium payload masses.
- Payload mass alone did not determine Falcon 9 landing success; orbit type and mission characteristics also influenced landing outcomes.

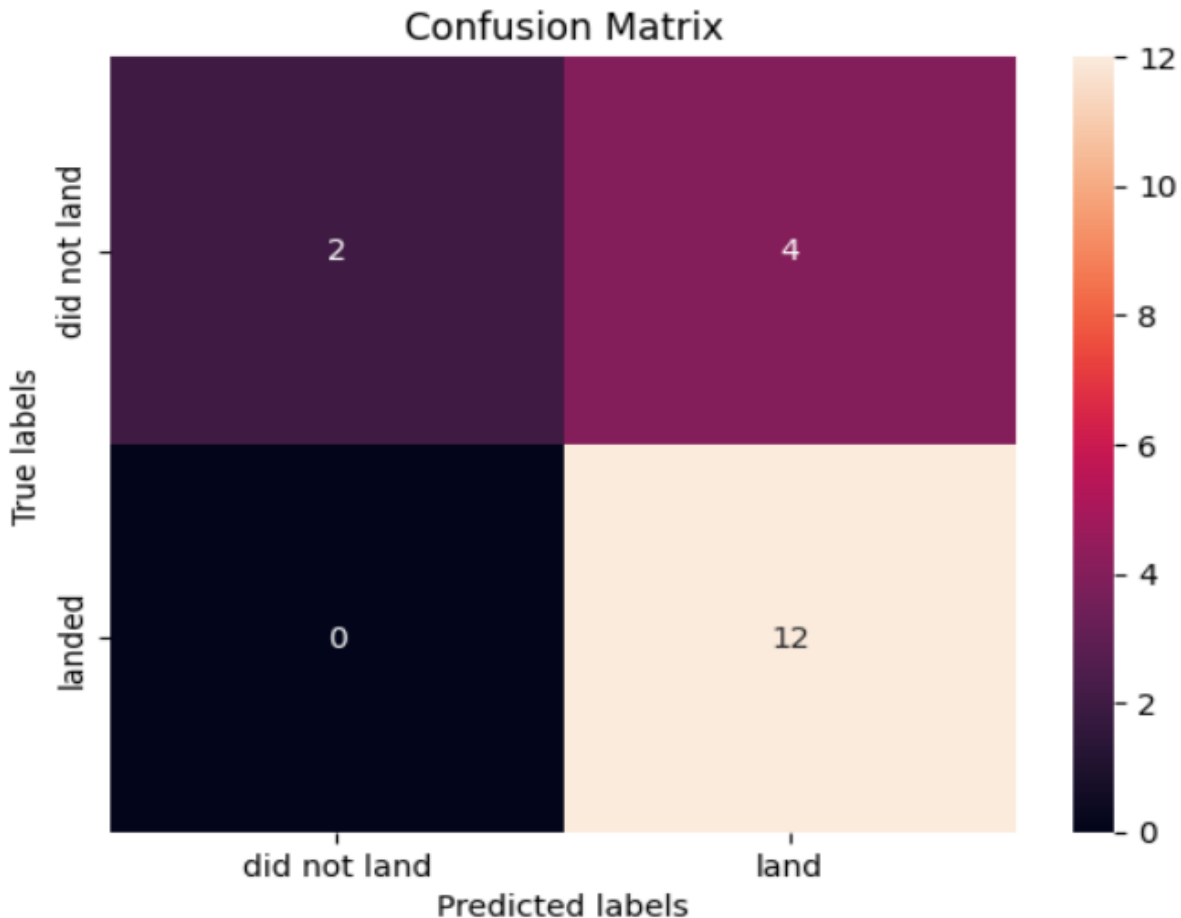
Yearly Launch Success Trend



• Key Findings

- Overall launch success rates improved over the years despite minor fluctuations.
- Success rates were very low during the early years (2010–2013).
- From 2016 onward, average launch success rates increased significantly.
- By 2019–2020, SpaceX consistently achieved high launch success rates, demonstrating improved booster recovery and mission reliability.

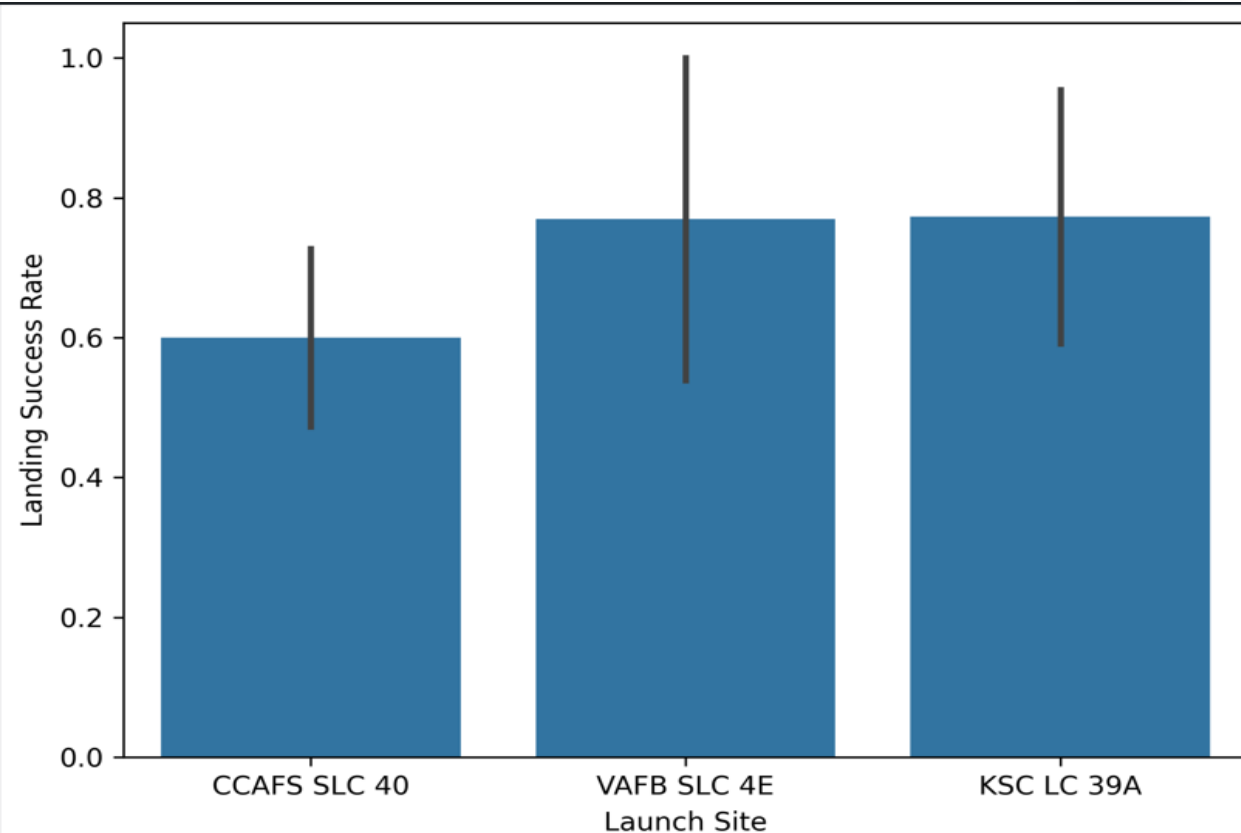
Model Evaluation – Confusion Matrix



- **Key Findings**

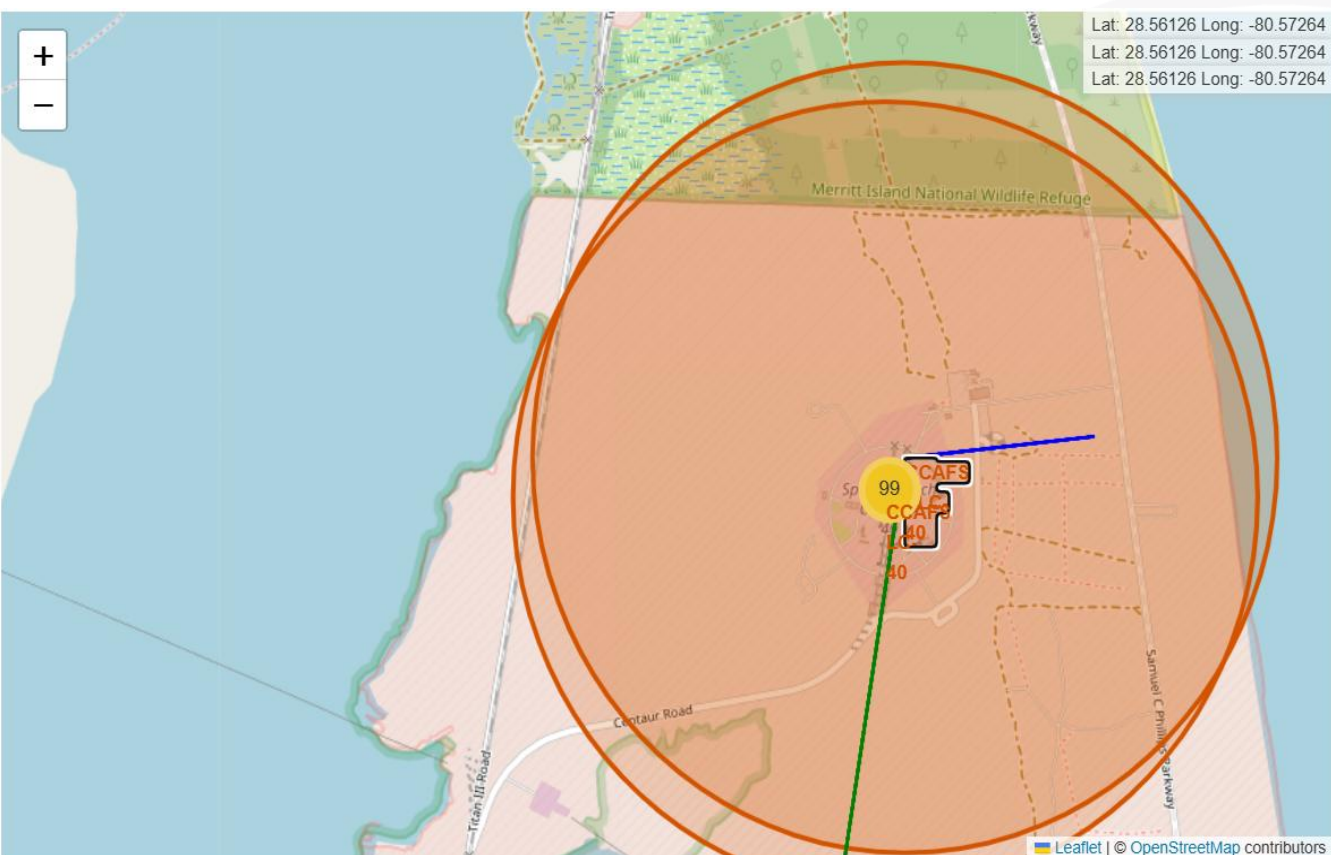
- The model achieved good predictive performance with no false negatives, although a few unsuccessful missions were classified as successful.
- **2 unsuccessful landings** were correctly classified (True Negatives).
- **4 unsuccessful landings** were incorrectly predicted as successful (False Positives).
- The model demonstrated strong performance in identifying successful landings, with no false negatives, although some unsuccessful missions were incorrectly classified as successful.

Landing Success by Launch Site



- **Key Findings**
- **KSC LC 39A** achieved the **highest average landing success rate** among all launch sites.
- **VAFB SLC 4E** also recorded a **high landing success rate**, indicating strong mission performance.
- **CCAFS SLC 40** showed the **lowest average landing success rate** in the dataset.
- Launch site significantly influenced Falcon 9 first-stage landing success.

Launch Site Proximity Analysis (CCAFS)



• Key Findings:

- CCAFS launch site is located close to the coastline.
- The coastal location supports safe launch trajectories over the ocean.
- The site is positioned away from major populated areas to improve safety.
- The proximity map highlights CCAFS's strategic coastal location for safe launch operations.

SPACEX FALCON 9 – FINDINGS & IMPLICATIONS

Findings

- Launch success rate improved over time.
- Launch site, payload mass, flight number, and orbit type significantly influenced landing success.
- The best-performing classification model demonstrated strong predictive performance.

Implications

- Machine learning can effectively predict Falcon 9 landing success.
- The predictive model can support mission planning and reduce launch costs.
- Data-driven insights help improve future SpaceX launch decisions.



Results – Visualizations & Dashboard

EDA Visualizations

- Flight Number vs. Launch Site
- Payload Mass vs. Launch Site
- Success Rate by Orbit
- Flight Number vs. Orbit
- Payload Mass vs. Orbit
- Yearly Launch Success Trend
- Landing Success by Launch Site

Interactive Dashboard

- SQL Query Results
- Folium Launch Site Map
- Plotly Dash Dashboard
- Confusion Matrix (Model Evaluation)

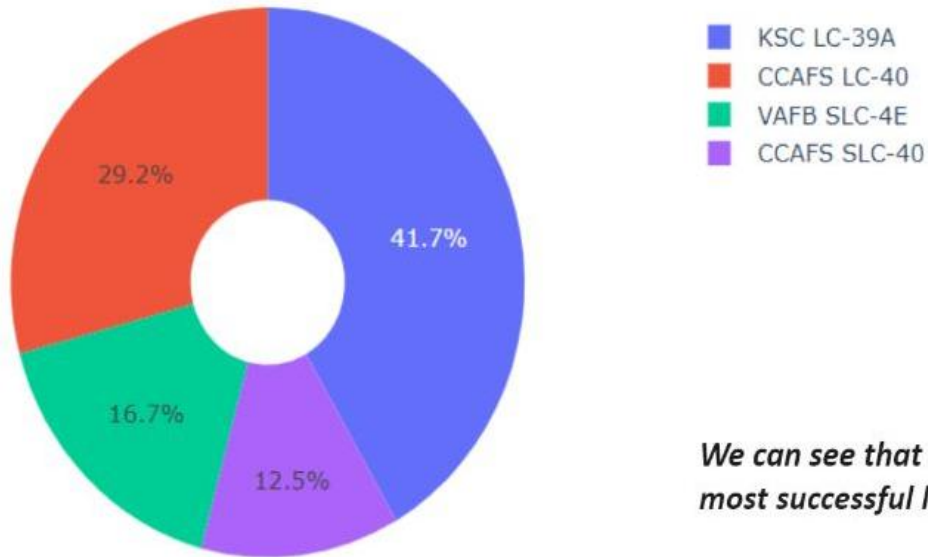
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Interactive Dashboard (Plotly Dash)

Total Success Launches by All Sites (Pie Chart)

Total Success Launches By all sites



We can see that KSC LC-39A had the most successful launches from all the sites

• Key Findings:

- Interactive dashboard developed using Plotly Dash.
- Displays successful launches across all launch sites.
- KSC LC-39A recorded the highest number of successful launches.

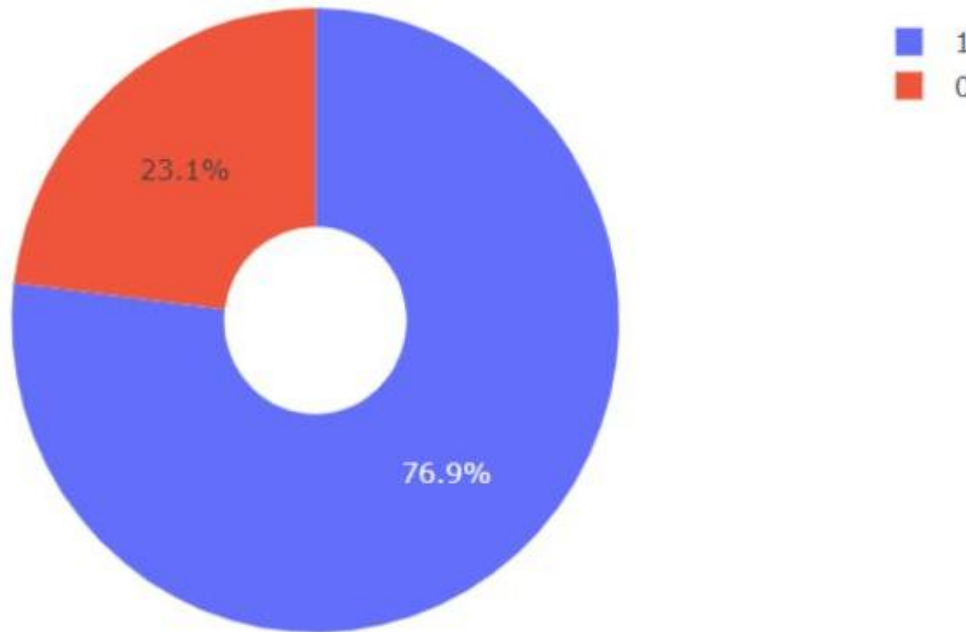
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Launch Site Analysis

KSC LC-39A: Success vs Failure



KSC LC-39A achieved a 76.9% success rate while getting a 23.1% failure rate

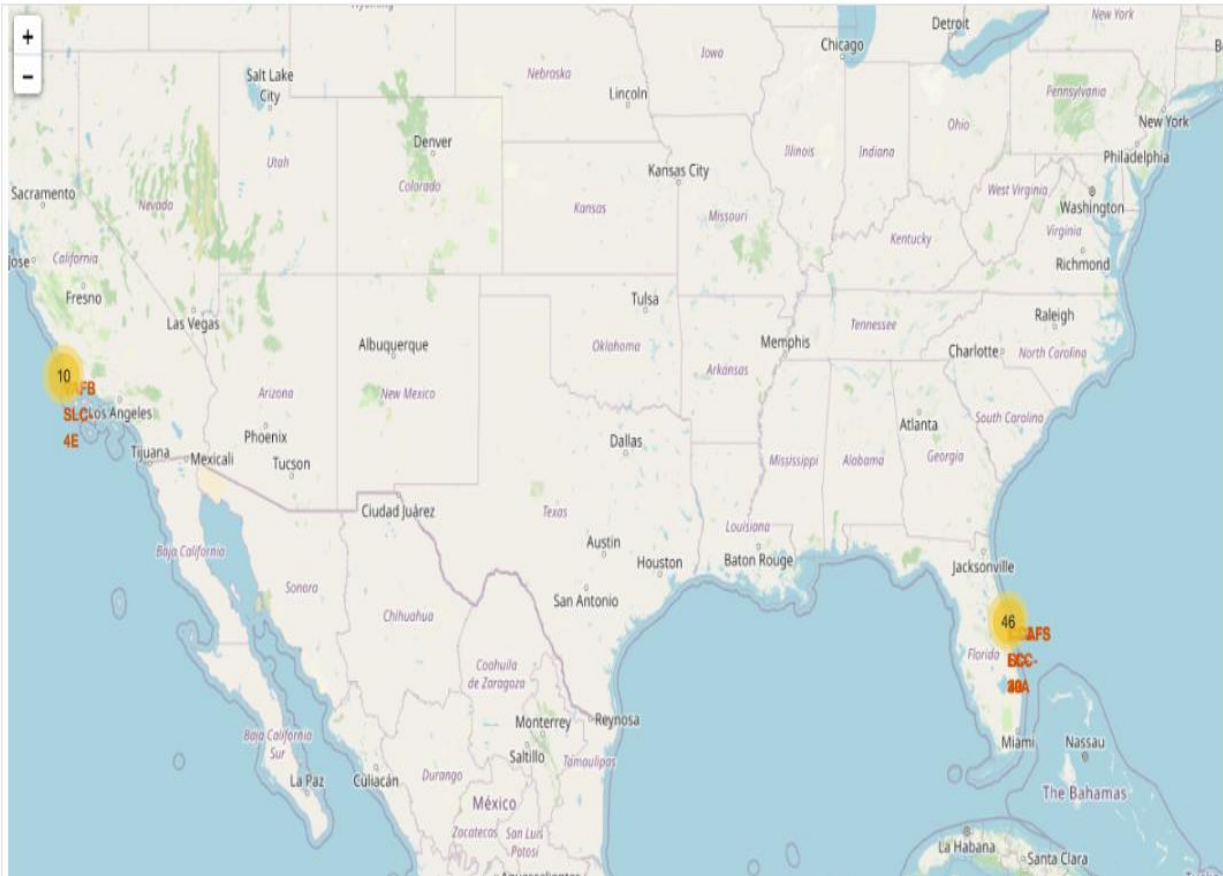
- **Key Findings:**
- Payload mass is compared with launch success.
- Different colors represent booster versions.
- The dashboard allows interactive payload range filtering.
- The visualization helps identify payload-related launch patterns.

GitHub Repository:

<https://github.com/Satish-KumarRam/IBM-Data-Science-Capstone-Project>

Interactive Launch Site Map (Folium)

Launch Site Distribution (Folium)

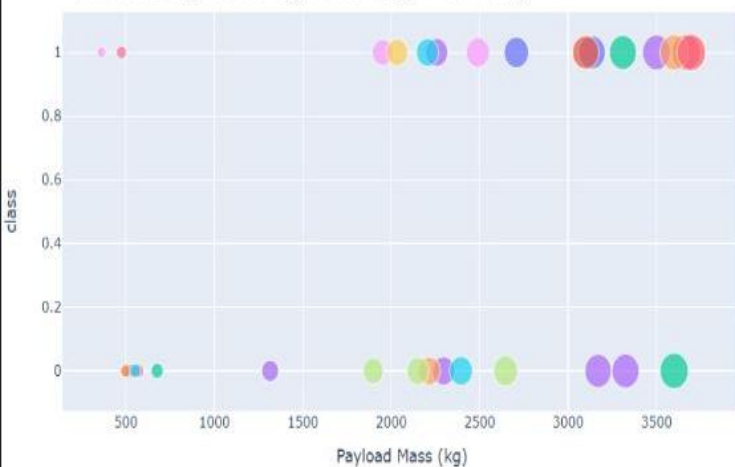


- **Key Findings:**
- Visualized Falcon 9 launch sites using Folium.
- Interactive markers display launch site locations and launch counts.
- CCAFS SLC-40 recorded the highest number of Falcon 9 launches, indicating it was the most active launch site.
- Geographic analysis helped compare launch site distribution.
- Location-based visualization helped identify launch site patterns and supported predictive analysis.

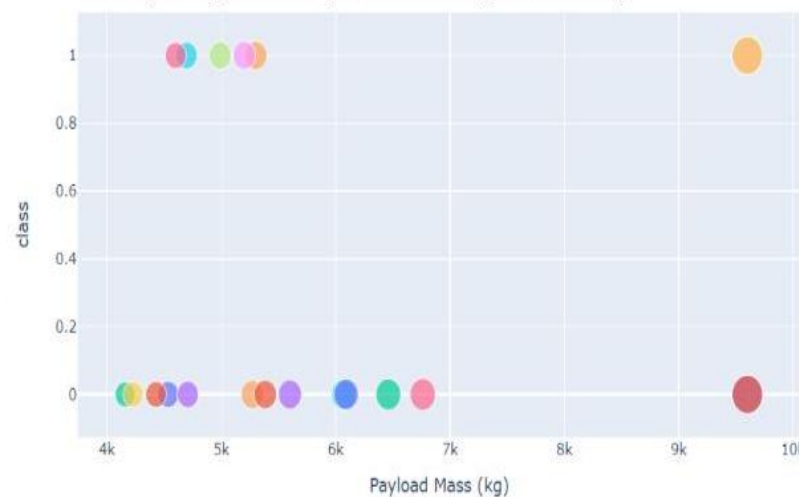
Payload Analysis

Payload vs Launch Success

Low Weighted Payload 0kg – 4000kg



Heavy Weighted Payload 4000kg – 10000kg



We can see the success rates for low weighted payloads is higher than the heavy weighted payloads

- **Key Findings:**
- Shows the relationship between payload mass and launch success.
- Interactive payload range slider enables dynamic filtering.
- Different booster versions are represented by different colors.

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Conclusion

• Key Takeaways

- The analysis identified several factors that influence **SpaceX Falcon 9 landing success**.
- **Flight Number** showed a positive relationship with landing success, indicating improved performance over time.
- Payload mass, orbit type, and launch site were key factors influencing landing success.

• Overall Findings

- Landing success rates improved significantly over time, reaching their highest levels by 2019 and remaining consistently high thereafter.
- Historical launch data provides valuable insights into Falcon 9 landing success and supports data-driven decision-making.

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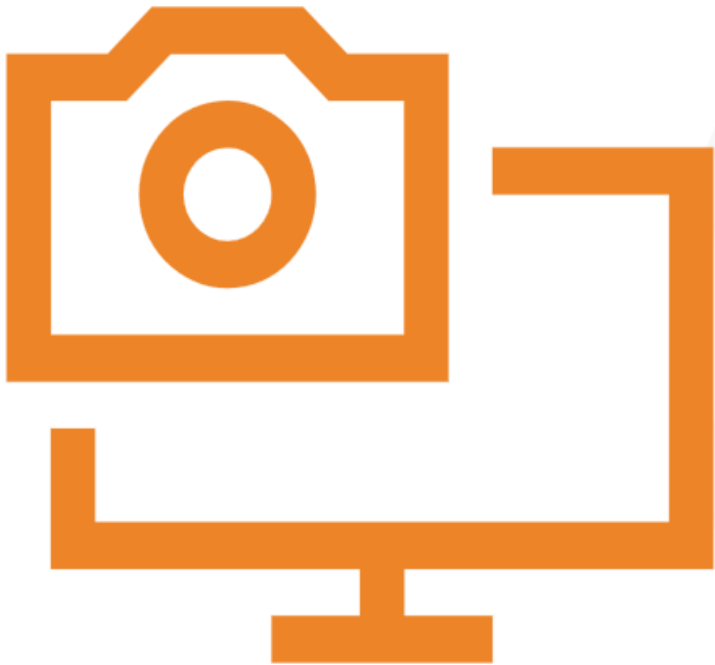
APPENDIX

- **Additional Analysis**

- SQL query outputs used for exploratory analysis.
- Interactive Plotly Dash dashboard for exploratory data analysis.
- Folium map showing launch site proximity.
- Additional EDA charts supporting model development.

GitHub Repository:

<https://github.com/Satish-KumarRam/IBM-Data-Science-Capstone-Project>



Thank You!

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